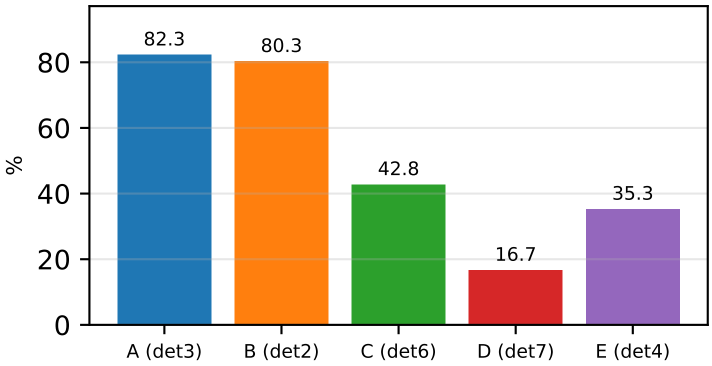
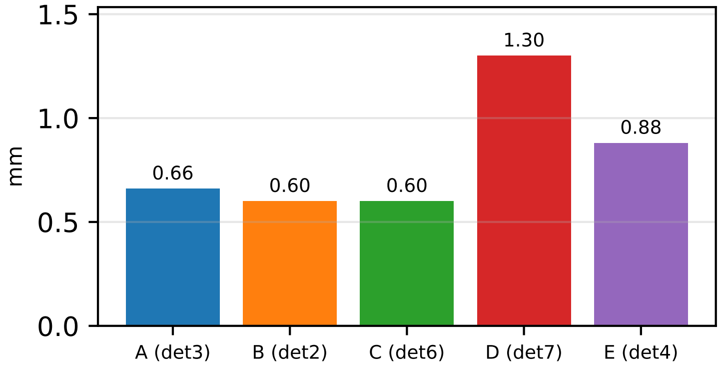
MX17 June 2026 cosmic-bench — fleet summary
M3 v2 reference tracking (NClus=4 & $\chi^2<1.0$); best long run per detector

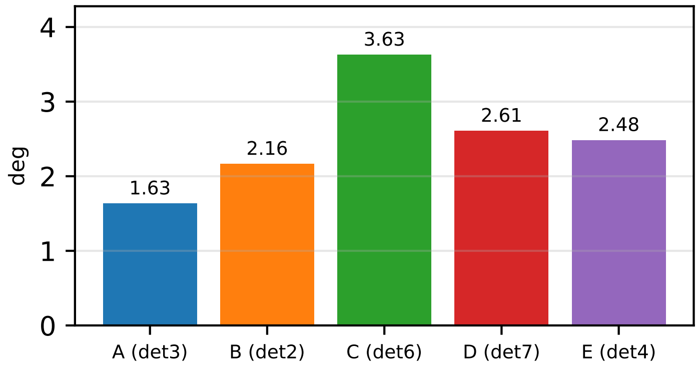
Best-run efficiency (within 5 mm)



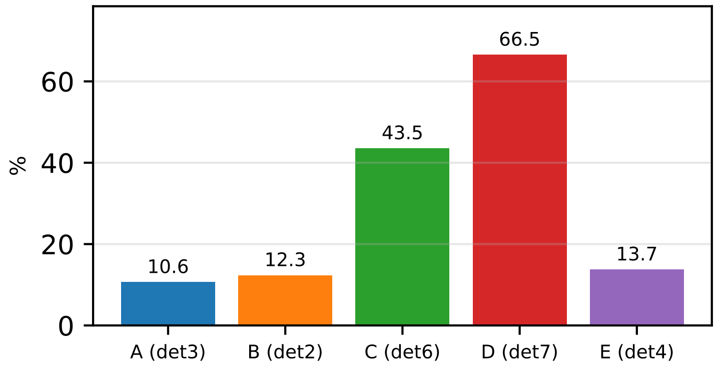
Spatial resolution (core σ)



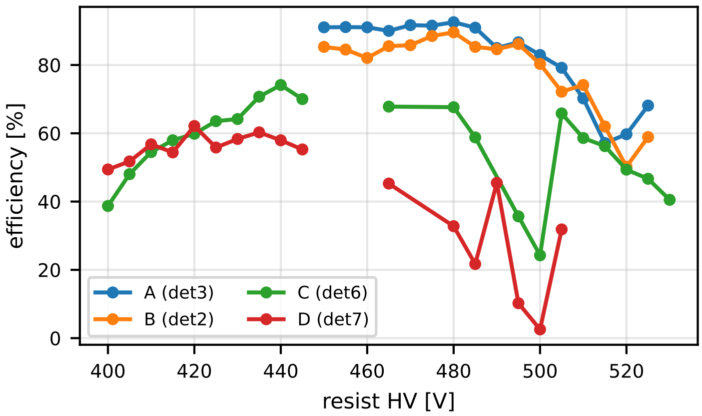
micro-TPC angular resolution (hybrid, |θ|<5° σ68)



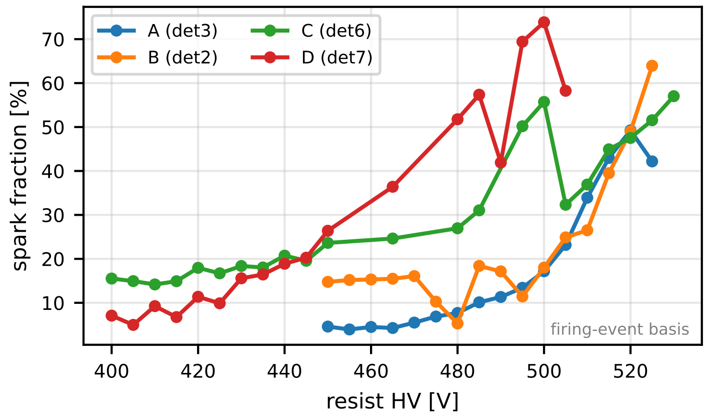
Spark rate (>50 strips)



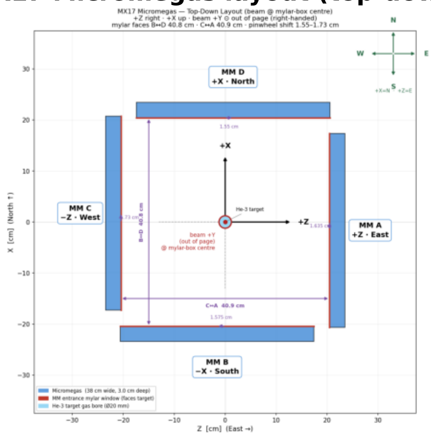
Efficiency vs resist HV



Spark rate vs resist HV



MX17 Micromegas layout (top-down)



Fleet summary

Det	Eff %	σ mm	θ°	Spark %
A (det3)	82.3	0.66	1.63	10.6
B (det2)	80.3	0.60	2.16	12.3
C (det6)	42.8	0.60	3.63	43.5
D (det7)	16.7	1.30	2.61	66.5
E (det4)	35.3	0.88	2.48	13.7

Generated 2026-07-25 · eff/σ/spark(bars) = best-stats run per detector, crossing-based spark · θ-res = hybrid σ68 |θ|<5° (script 34, holdout) (≠ eff run for det3)

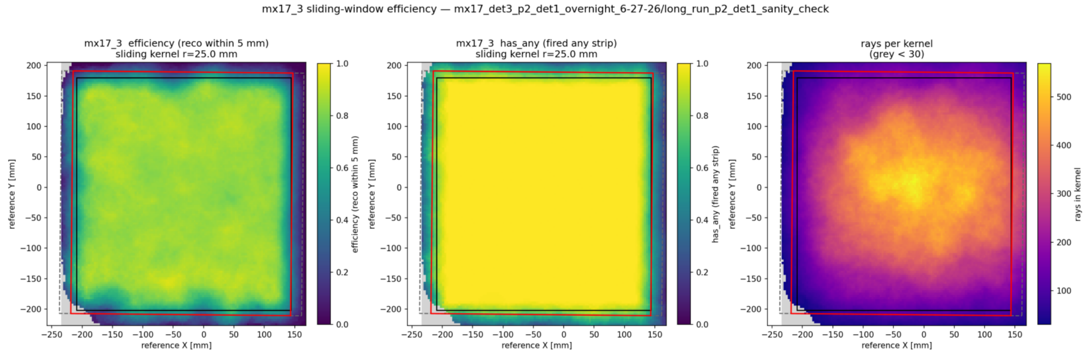
Detector A

Resist 490 V Drift 1000 V

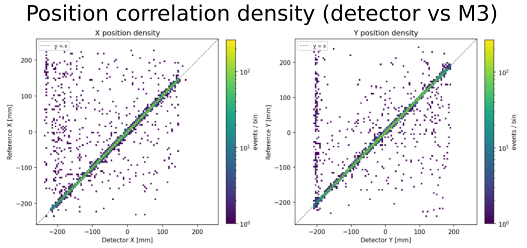
```
Detector A  (mx17_3)
mx17_det3_p2_det1_overnight_6-27-26
subrun: long_run_p2_det1_sanity_check
FEU X/Y: 7/8    z: 702 mm
align: θ=89.45°  z=714.0 mm
clean M3 rays: 22417
median |r|: 1.03 mm
loss: hit-no-reco 0.1%  silent 0.5%
```

82.3%	0.66 mm	99.5%	88.8%	10.6%
Efficiency (≤5 mm)	Resolution (core σ)	Fired any strip	Reconstructed	Spark rate (>50 strips)

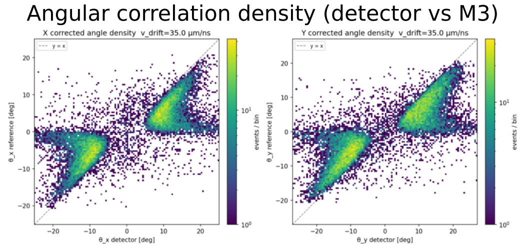
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



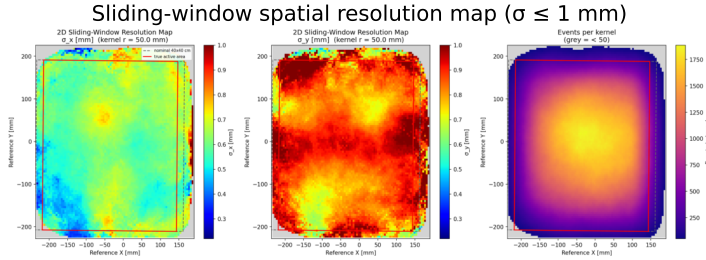
Position correlation density (detector vs M3)



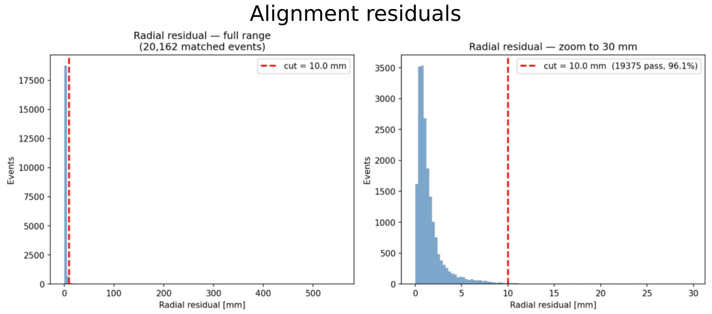
Angular correlation density (detector vs M3)



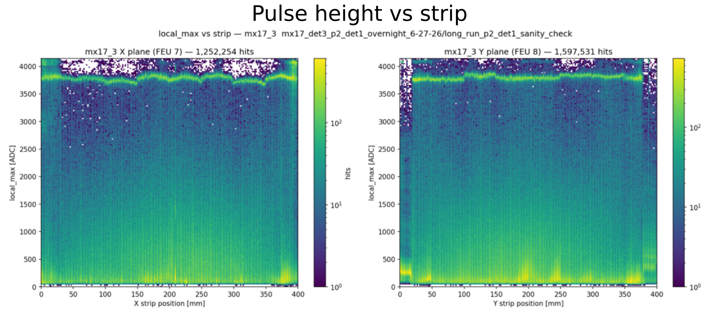
Sliding-window spatial resolution map (σ ≤ 1 mm)



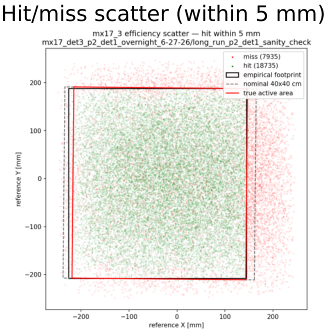
Alignment residuals



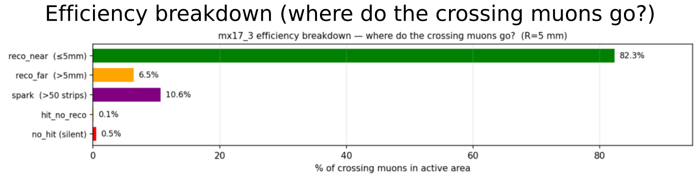
Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



```
Detector B  (mx17_2)
mx17_det2_det3_overnight_6-22-26
subrun: longer_run
FEU X/Y: 6/8    z: 702 mm
align: θ=89.15°  z=713.0 mm
clean M3 rays: 3772
median |r|: 0.96 mm
loss: hit-no-reco 0.1%  silent 0.8%
```

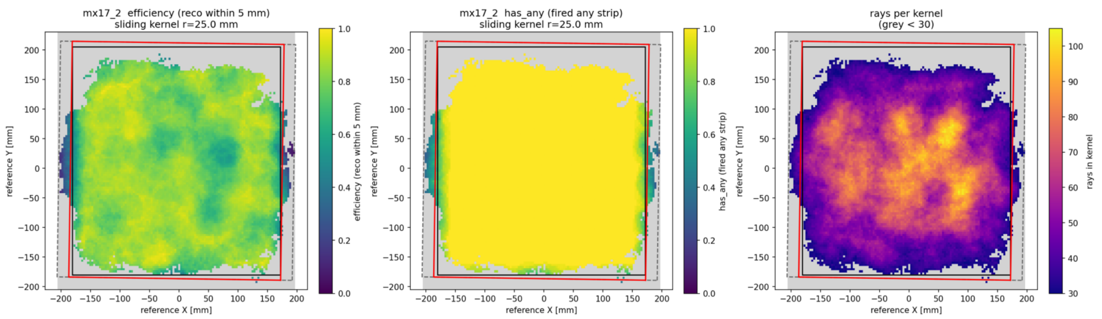
Detector B

Resist 495 V   Drift 1000 V

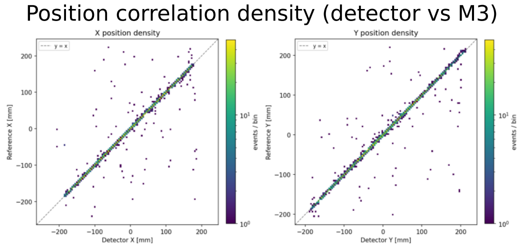
80.3%	**0.60 mm**	**99.2%**	**86.7%**	**12.3%**
Efficiency (≤5 mm)	Resolution (core σ)	Fired any strip	Reconstructed	Spark rate (>50 strips)

Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)

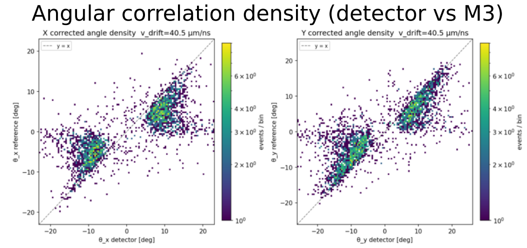
mx17_2 sliding-window efficiency — mx17_det2_det3_overnight_6-22-26/longer_run



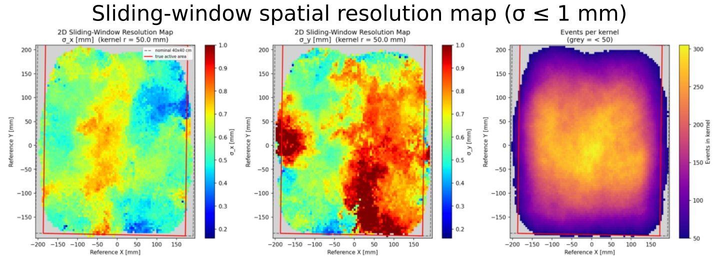
Position correlation density (detector vs M3)



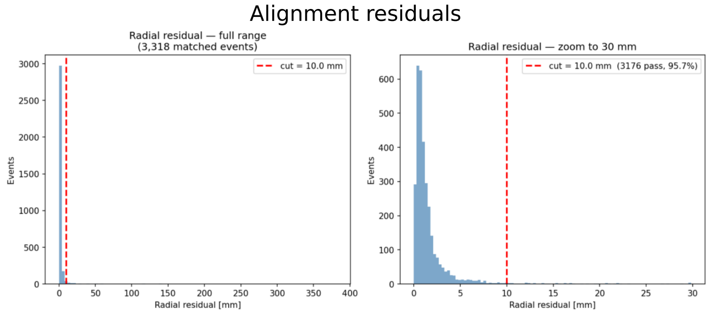
Angular correlation density (detector vs M3)



Sliding-window spatial resolution map (σ ≤ 1 mm)

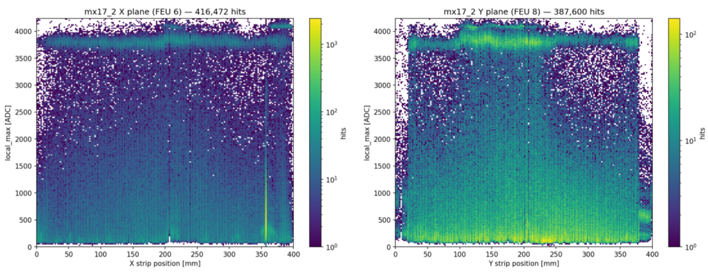


Alignment residuals



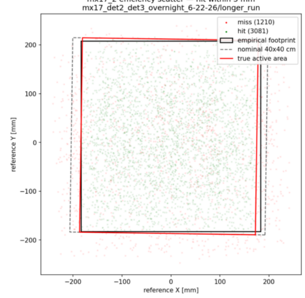
Pulse height vs strip

local_max vs strip — mx17_2 mx17_det2_det3_overnight_6-22-26/longer_run

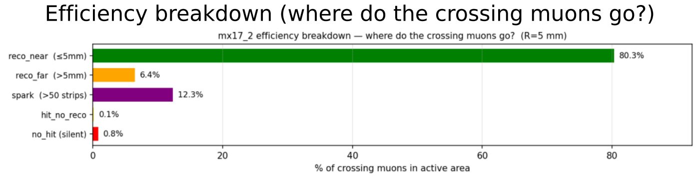


Hit/miss scatter (within 5 mm)

mx17_2 efficiency scatter — hit within 5 mm
mx17_det2_det3_overnight_6-22-26/longer_run



Efficiency breakdown (where do the crossing muons go?)



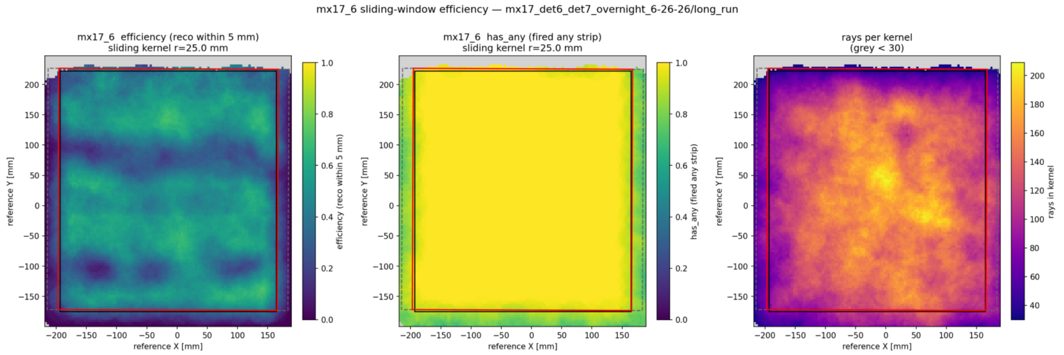
Detector C

```
Detector C  (mx17_6)
mx17_det6_det7_overnight_6-26-26
subrun: long_run
FEU X/Y: 3/4    z: 232 mm
align: θ=89.75°  z=243.0 mm
clean M3 rays: 10366
median |r|: 1.03 mm
loss: hit-no-reco 4.4%  silent 1.4%
```

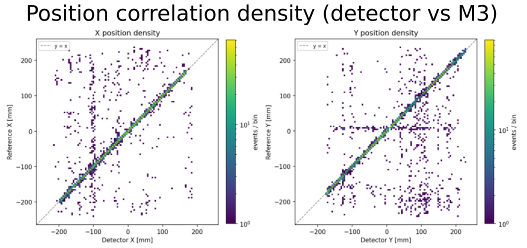
Resist 495 V   Drift 700 V

42.8%	0.60 mm	98.6%	50.8%	43.5%
Efficiency (≤5 mm)	Resolution (core σ)	Fired any strip	Reconstructed	Spark rate (>50 strips)

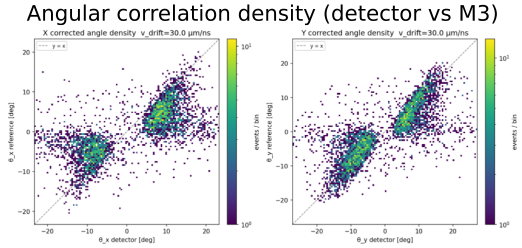
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



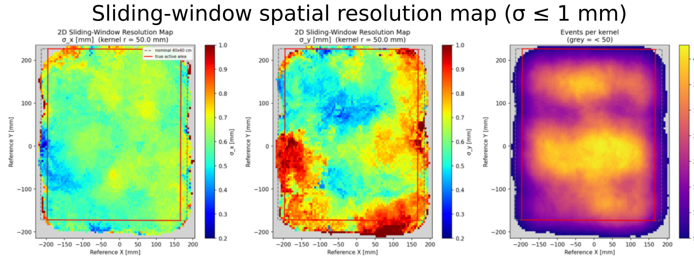
Position correlation density (detector vs M3)



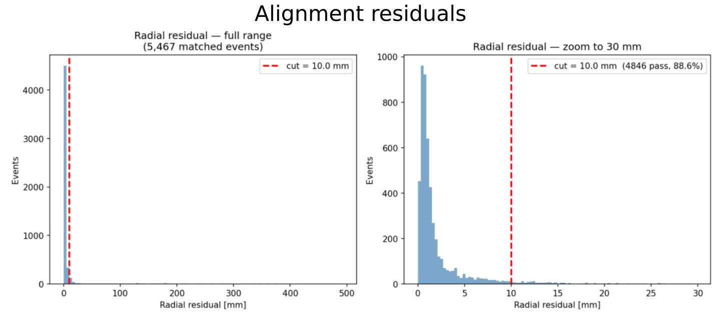
Angular correlation density (detector vs M3)



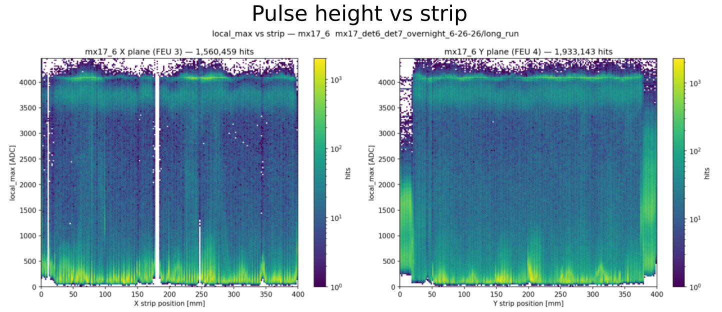
Sliding-window spatial resolution map (σ ≤ 1 mm)



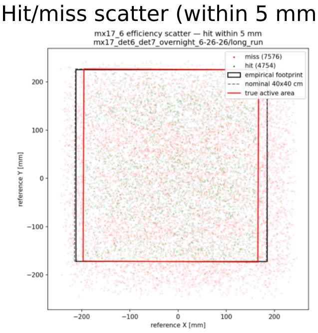
Alignment residuals



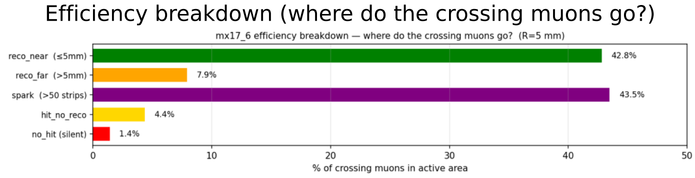
Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



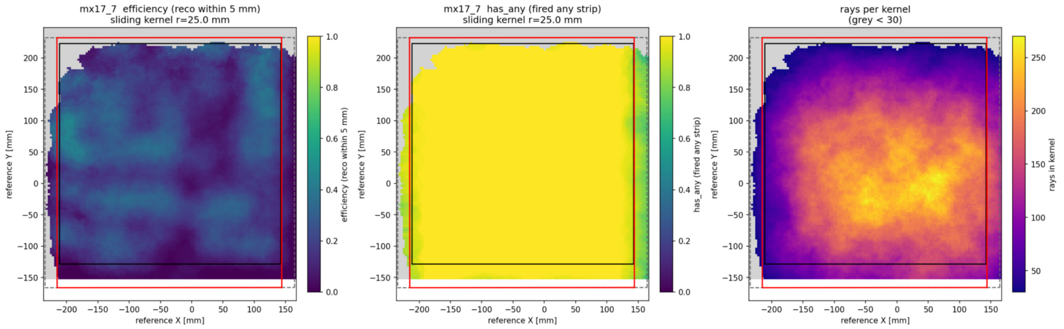
Detector D

```
Detector D  (mx17_7)
mx17_det6_det7_overnight_6-26-26
subrun: long_run
FEU X/Y: 6/8    z: 702 mm
align: θ=90.05°  z=712.0 mm
clean M3 rays: 10651
median |r|: 2.33 mm
loss: hit-no-reco 5.5%  silent 1.1%
```

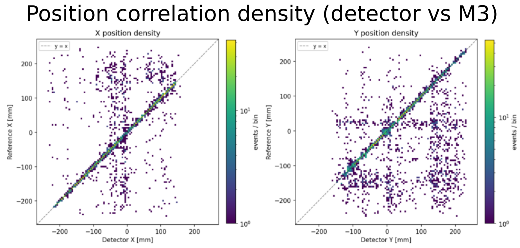
Resist 495 V   Drift 700 V

16.7%	1.30 mm	98.9%	26.9%	66.5%
Efficiency (≤5 mm)	Resolution (core σ)	Fired any strip	Reconstructed	Spark rate (>50 strips)

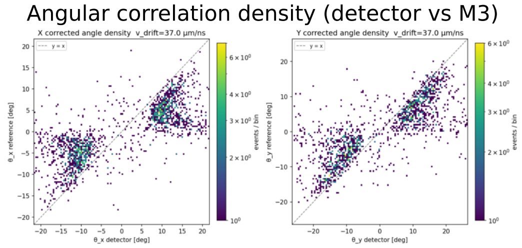
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



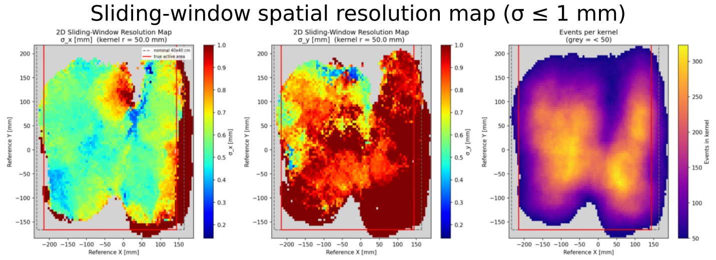
Position correlation density (detector vs M3)



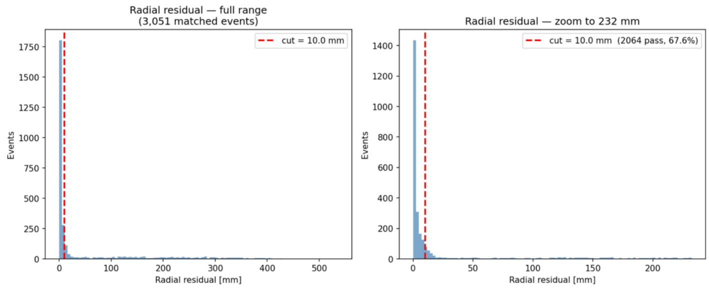
Angular correlation density (detector vs M3)



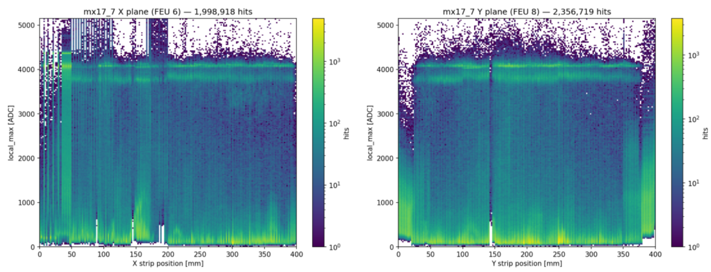
Sliding-window spatial resolution map (σ ≤ 1 mm)



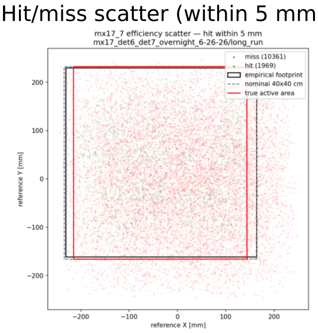
Alignment residuals



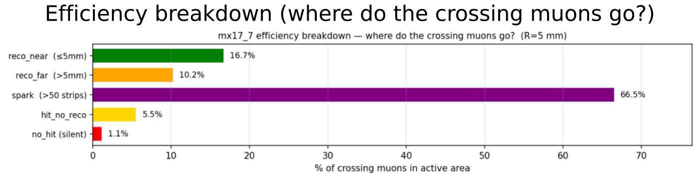
Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



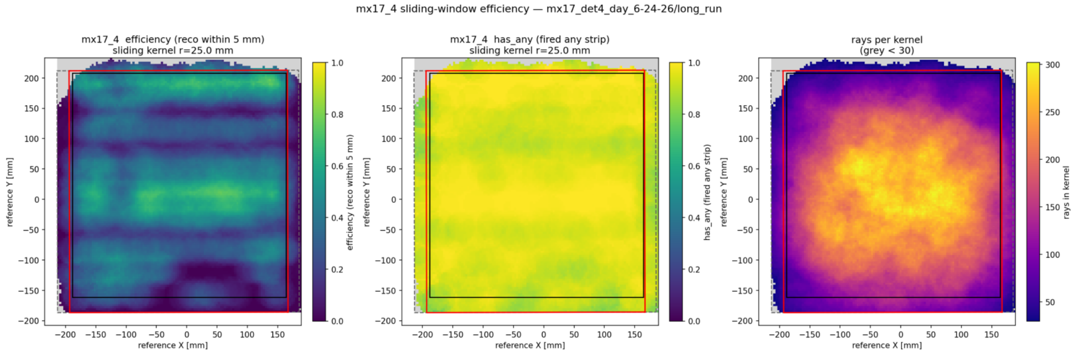
Detector E

Resist 495 V   Drift 600 V

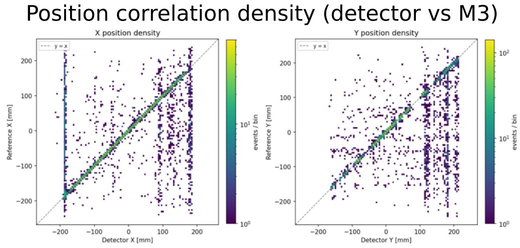
```
Detector E  (mx17_4)
mx17_det4_day_6-24-26
subrun: long_run
FEU X/Y: 6/8    z: 702 mm
align: θ=90.15°  z=712.0 mm
clean M3 rays: 12570
median |r|: 1.50 mm
loss: hit-no-reco 37.3%  silent 4.4%
```

35.3%	0.88 mm	95.6%	44.7%	13.7%
Efficiency (≤5 mm)	Resolution (core σ)	Fired any strip	Reconstructed	Spark rate (>50 strips)

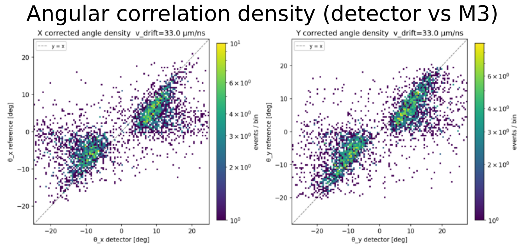
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



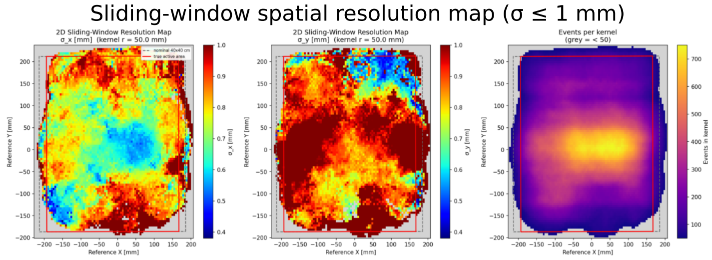
Position correlation density (detector vs M3)



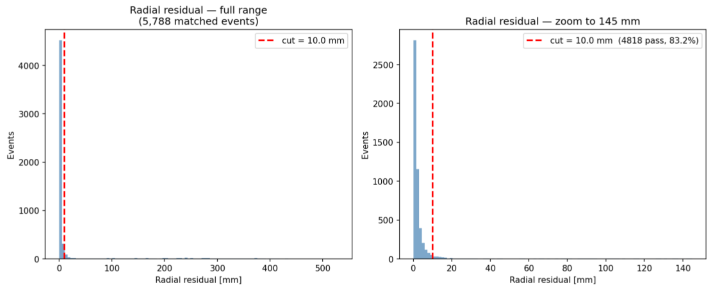
Angular correlation density (detector vs M3)



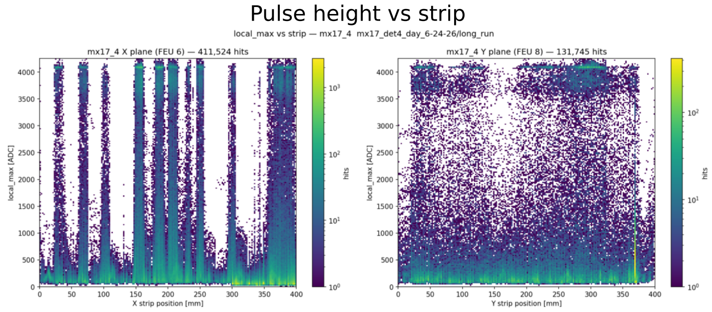
Sliding-window spatial resolution map (σ ≤ 1 mm)



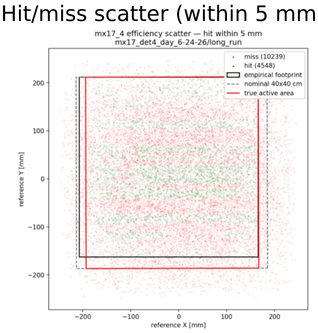
Alignment residuals



Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)